

PROFESSIONAL PROFILE

Applied Machine Learning Engineer and Data Scientist with a ten-year track record spanning commercial telecommunications, physical hardware R&D and advanced artificial intelligence. Combines a rigorous academic foundation with practical experience deploying deep learning inference engines onto edge hardware. Proven expertise in building multi-terabyte data pipelines and engineering Transformer models to solve physical-world challenges. Experience grounded in strong domain knowledge of 5G infrastructure, IoT communications and RF propagation, validated by commercial intelligence roles and patented hardware circuit design. A capable technical mentor and communicator, focused on translating complex mathematical theories into robust, production-ready solutions for the telecommunications, deep tech and defence sectors.

TECHNICAL PROFICIENCIES

Machine Learning & AI: PyTorch, Transformers (Sequence Modelling, Self-Supervised Pretraining), Neural Networks, Ordinal Classification, Time-Series Analysis

Data Engineering & Computing: Python, C, MATLAB, Data Pipeline Architecture, Feature Engineering, Reproducible Model Evaluation

Edge & Cloud Deployment: NVIDIA A100 (Brev Compute), NVIDIA Jetson AGX Orin, TensorRT

Telecommunications & RF: LTE/5G Baseband Capture, Software-Defined Radio (SDR), USRP, CST, HFSS, Simulink, MATLAB

Infrastructure & Version Control: Linux, Git, VS Code (Remote/SSH)

EDUCATION

Doctor of Philosophy (Ph.D.), Electrical Engineering

The University of Sydney, Australia | 2023

Master of Science (M.Sc.), Electrical Engineering

Korea Advanced Institute of Science and Technology (KAIST), Republic of Korea | 2016

Bachelor of Science (B.Sc.), Electrical Engineering

Iran University of Science and Technology (IUST), Iran | 2014

PROFESSIONAL SKILLS

Applied Research & Development

Hardware-Software Integration

Predictive Analytics

Algorithm Optimisation

Experimental Design & Validation

Technical Mentorship

Telecommunications Strategy

Project Delivery

Stakeholder Communication

Innovation & IP Generation

CORE COMPETENCIES

- Proven ability to translate physical-layer telecommunications data into machine learning models for real-world deployment.
- Experienced managing data lifecycle, from theoretical algorithm design and testing through to edge hardware integration.
- Strong background in engineering and processing, multi-terabyte datasets derived from live, high-noise environments.
- Demonstrated capacity to transform experimental research into viable R&D outputs, including patented physical hardware.
- Background in technical instruction, effectively communicating mathematical and engineering concepts to diverse cohorts.
- Advanced understanding of wireless network dynamics, utilising statistical modelling to drive operational intelligence and performance improvements.

EMPLOYMENT HISTORY / PROJECTS

University of Technology Sydney, Sydney | Aug 2023 – Present

Postdoctoral Data Scientist

Directed applied machine learning projects in partnership with NVIDIA, TPG Telecom and the NSW Department of Planning. Built and deployed real-time environmental sensing systems that repurposed existing 4G/5G telecommunications networks into predictive intelligence tools for flood and storm management.

- Deployed an INT8 TensorRT inference engine onto NVIDIA Jetson AGX Orin edge hardware to process real-time estimations and stream data to a live dashboard.
- Trained Transformer networks using self-supervised pretraining on NVIDIA A100 cloud compute clusters to evaluate 5G channel measurements.
- Architected scalable data pipelines to aggregate, clean and process over 34 TB of combined physics-based synthetic data and multi-frequency field captures.
- Engineered Transformer-based sequence and ordinal classifiers using high-dimensional 4G/5G signals, radio power metadata and delay-Doppler feature extraction.
- Constructed rigorous, day-based chronological train/validation/test evaluation frameworks incorporating post-training calibration to guarantee model reliability across temporal shifts.
- Programmed end-to-end signal processing workflows to execute LTE decoding, extract baseband metrics and apply wavelet-based time-frequency analysis.
- Executed large-scale physical field trials utilizing 4-antenna arrays and software-defined radio (SDR) receivers to capture live baseband I/Q datasets.

The University of Sydney, Sydney | Oct 2019 – Oct 2025

Casual Academic (Technical Instructor – Machine Learning & RF)

Delivered advanced technical instruction across physical-layer telecommunications and applied machine learning.

- Mentored technical cohorts in applied machine learning, instructing on the architecture, training and deployment of predictive models using PyTorch, TensorFlow and Scikit-learn.
- Directed laboratory execution for IoT critical infrastructure, guiding students through the physical integration of edge computing hardware (Raspberry Pi) with real-time data platforms (ThingSpeak).
- Instructed classes on predictive analytics and foundational data science mechanics, teaching robust data manipulation, pipeline construction and statistical modelling using Pandas and NumPy.
- Translated theoretical RF propagation and antenna design into practical engineering simulations, training students on the application of CST, HFSS and MATLAB for physical-layer network analysis.
- Taught complex mathematical frameworks for telecommunications, focusing on the implementation of BCH, Turbo, Convolutional and LDPC error control coding to ensure network signal integrity.

The University of Sydney, Sydney | Oct 2019 – Oct 2023

Doctoral Researcher (Machine Learning & 5G IoT)

Engineered and evaluated machine learning algorithms designed to optimise 5G IoT communications and physical-layer wireless performance.

- Engineered an unsupervised learning framework using Gaussian Mixture Models (GMM) to autonomously detect, isolate and process signals from 5G IoT devices.
- Built distributed machine learning models to operate in high-noise RF environments, deploying Convolutional Neural Networks (CNNs) to maintain classification accuracy despite physical signal degradation.
- Analysed physical-layer wireless system performance by applying statistical modelling and unsupervised algorithms to identify transmission inefficiencies.
- Modelled RF propagation and antenna configurations using CST, HFSS and MATLAB to establish baseline physical constraints for communication network simulations.

Future Wave Ultra Tech | 2018 – 2019

Telecommunication Engineer (Data & Analytics)

Designed and implemented automated operational intelligence tools for commercial mobile networks, translating raw, multi-vendor telecommunications data into actionable business metrics.

- Created an automated mobile-network analytics platform, building robust ETL pipelines to extract, clean and aggregate data from complex, multi-vendor Operational Support Systems (OSS).
- Programmed custom analytical logic to compute network KPIs and detect live system events, with focus on establishing reliable alarm monitoring for commercial network infrastructure.
- Generated automated optimisation reports based on large-scale data aggregation, directly driving operational decision-making and physical network improvements.

Korea Advanced Institute of Science and Technology (KAIST) | 2014 – 2016

Hardware R&D Engineer (Wireless Systems)

Focused on the physical-layer improvement of wireless power transfer, resulting in patented hardware circuitry and optimised algorithmic performance.

- Designed and patented novel hardware circuit architecture that improved physical efficiency of wireless charging systems.
- Developed and validated two optimisation algorithms designed to reduce charging times across operational constraints.
- Quantified physical charging performance against distance variables by constructing advanced computational models in MATLAB and Simulink.

HONOURS & AWARDS

NVIDIA Data Science Grant (2026)

Paulette Isabel Jones Scholarship (2023)

University of Sydney Postgraduate Research Support Scholarship (2022 – 2023)

The University of Sydney Postgraduate Scholarship (Ph.D.) (2019 – 2023)

KAIST Postgraduate Scholarship (M.Sc.) (2014 – 2016)

PERSONAL ATTRIBUTES

- Pragmatic approach to solving engineering problems at the intersection of physical hardware and software logic.
- High adaptability and resilience when operating within unstructured or ambiguous research-driven environments.
- A highly analytical mindset focused on extracting practical, actionable value from real-world data.
- Clear communication style, capable of translating technical information for cross-functional teams and external partners.
- Uncompromising commitment to rigorous, evidence-based testing and experimental reproducibility.
- Ownership and accountability for delivering on project milestones, from initial data capture to final deployment.

LANGUAGES

English (Bilingual Proficiency) • Persian (Native) • Arabic (Elementary)

REFERENCES

Andrew Zhang

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Dr. Reza Ahmadi

Senior Machine Learning Engineer, Geobotica

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